Structural Equation Modeling with Onyx

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Preface

This is a self-paced workshop for Onyx, a graphical user-interface for structural equation modeling.

![images/somemodels.png]

We assume that you already familiar with the basics of Structural Equation Modeling.

The workshop has some introductory chapters that cover basic workflow elements in Onyx, such as basics of model specification, evaluating model fit, or specifying multiple group models.

1 Introduction

Onyx's interface is somewhat unusual as it presents the user with an empty desktop. All major interactions in Onyx are via context-menus or left- and right-mouseclicks that provide context-dependent shortcuts.

This workshop is work-in-progress and has several chapters targeted at readers that are familiar with Structural Equation Modeling and want to learn about how to best use Onyx in their scientific work.

There is no need to follow the order of the modules as they are presented. Pick a topic about which you would like to learn more. The topics contain some basic information, some demonstrations, and practical exercises for you to follow along.

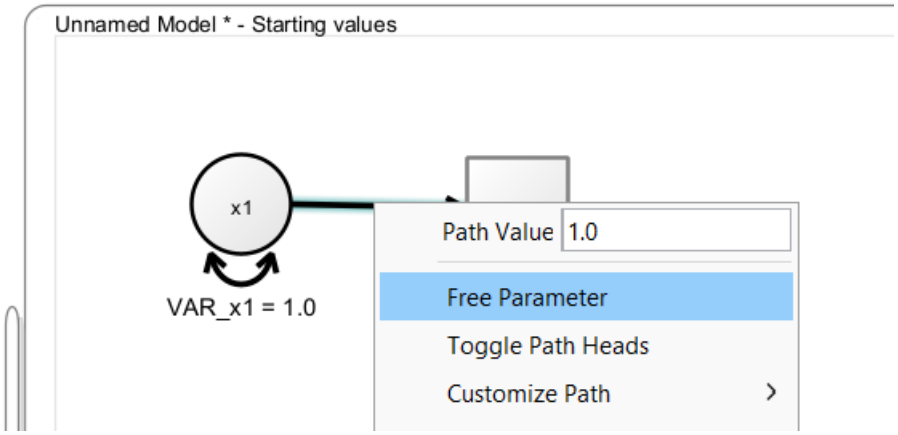
2 Basic Modeling

To create a model, either right-click on the empty desktop and select “Create new Model -> Create Empty Model” or double-click on the desktop. This creates an empty model view

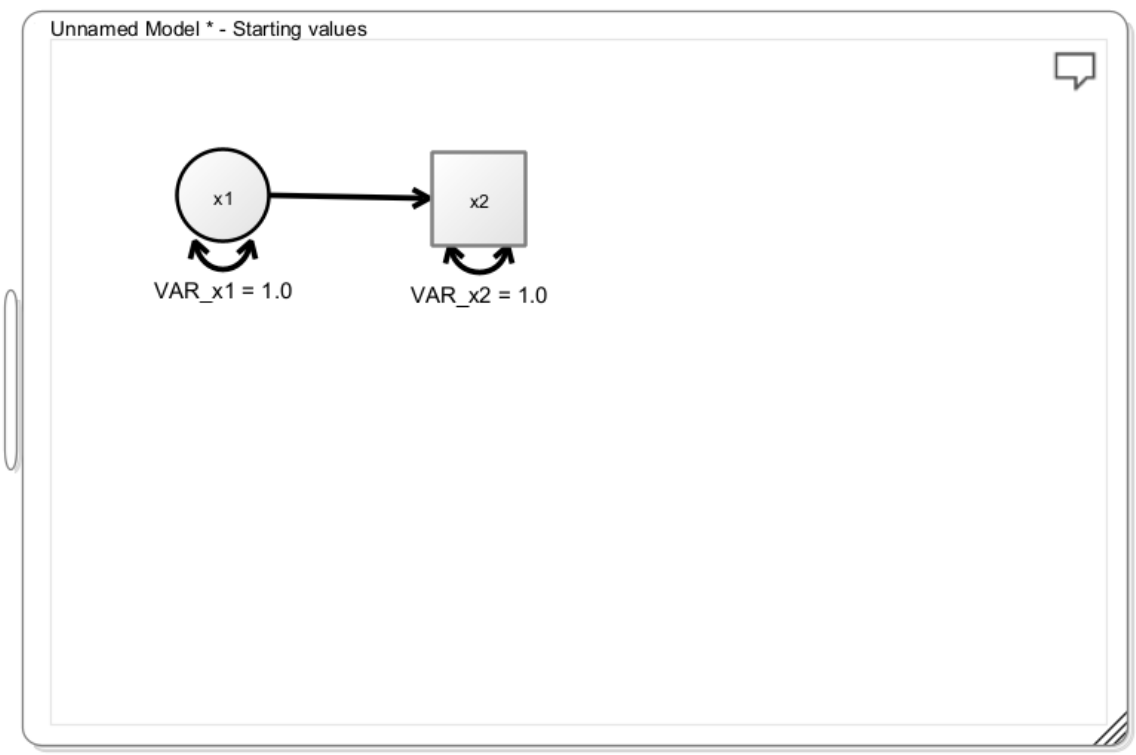
You can now add variables, by either right-clicking on the empty view and selecting “Create Variable”. Then you can either add a latent variable (depicted as a circle), an observed variable (depicted as a rectangle), a constant variable (depicted as a triangle). Beyond standard RAM notation , you can also add a multiplication node that allows you to model latent interactions. As a short-cut, you can simply double-click on an empty space in the model, to create a latent variable or double-click while holding down the SHIFT-key to add an observed variable.

To add a path between two variables, press the right mouse button on a variable, hold it down while dragging the path on to the target variable, then release the mouse button. This creates a regression path with a value fixed to “1” between the variables. When you hold down SHIFT

while doing so, you create a covariance path. Right-click the path and select “Free Parameter” if you want this path to be freely estimated.



Try to add a latent and an observed variable and connect them with a path from the latent to the observed variable, and your model should look like this:

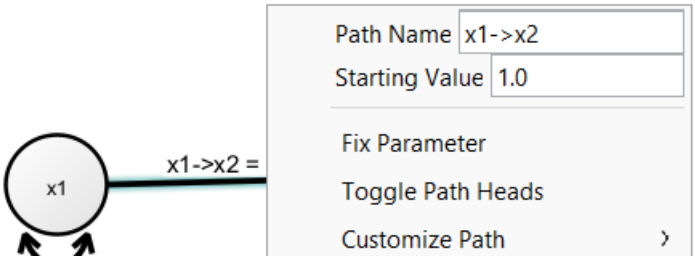


2.1 Changing Visuals

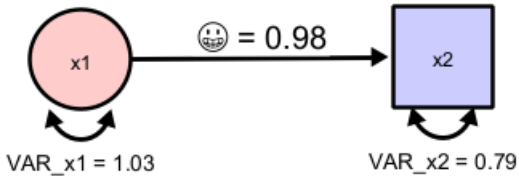
Onyx has a variety of options to adjust the appearance of the graph. Right-click on a node and choose "Customize Variable" to explore various options to adjust font size, line colors, fill colors, etc. The same is possible if you choose "Customize Path" when you right-click on a path.

2.2 Renaming Parameters

If you free a parameter on a path (representing either a variance, covariance, a regression, or a mean), you can change the name of the parameter by right-clicking on the path and then changing the "Path Name". By default, these parameters are named based on the adjacent variables but you can choose your own name here.



Note that Onyx has Unicode support, that is, you can directly use unicode characters such as letters from the greek alphabet. Thus, it would even be possible to use emojis as parameter labels even though I do not recommend that:

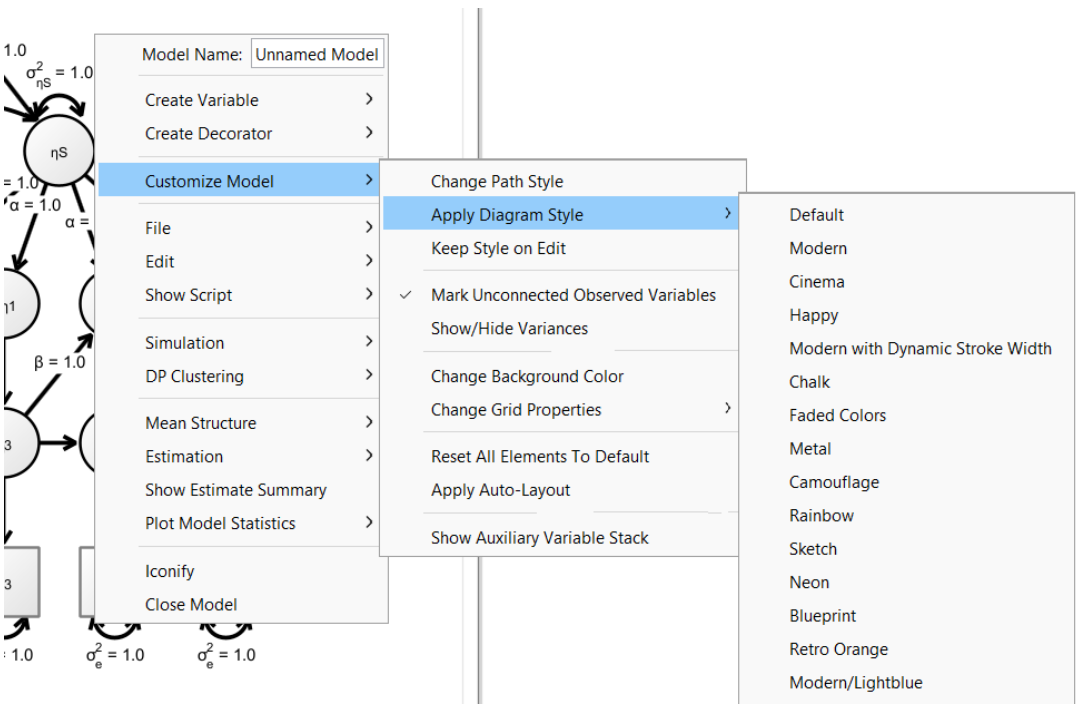


Onyx further supports a very limited subset of LaTeX commands to render parameter labels. This pseudo LaTeX input is triggered by starting labels with a dollar sign. This pseudo input supports lowercase and uppercase greek letters, as well as subscript and superscripts. Here are some examples that are valid parameter names in Onyx:

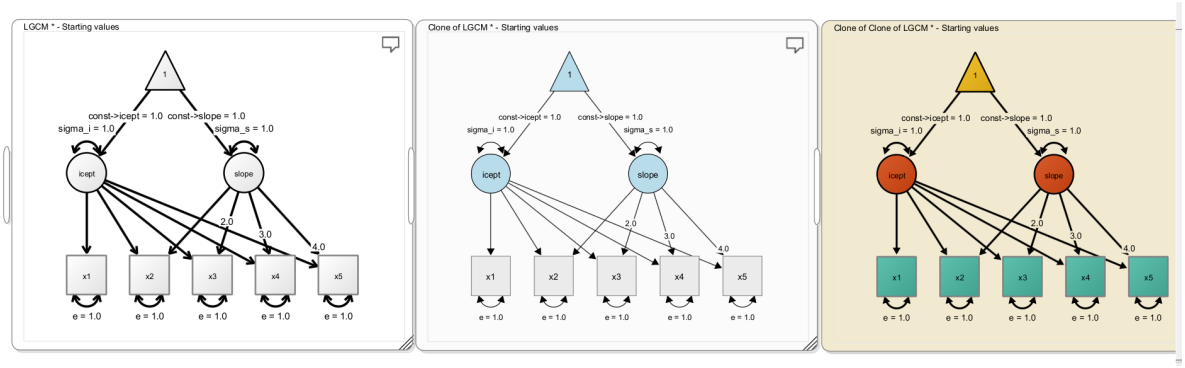
Input	Output
	α
^2	β^2
_{person}	γ_{person}
^2_i	Ω_i^2

2.3 Diagram Styles

Onyx ships some pre-made styles for your graphs. Choose "Customize Model -> Apply Diagram Style" and then choose from the available choices. Certainly, not all of them are useful for scientific communication but you may want to start with Modern or Modern/Lightblue. Further, you may want to choose "Keep Style on Edit", which continues to automatically apply the chosen style to newly added variables or paths in the diagram (otherwise, they are added using the default style).

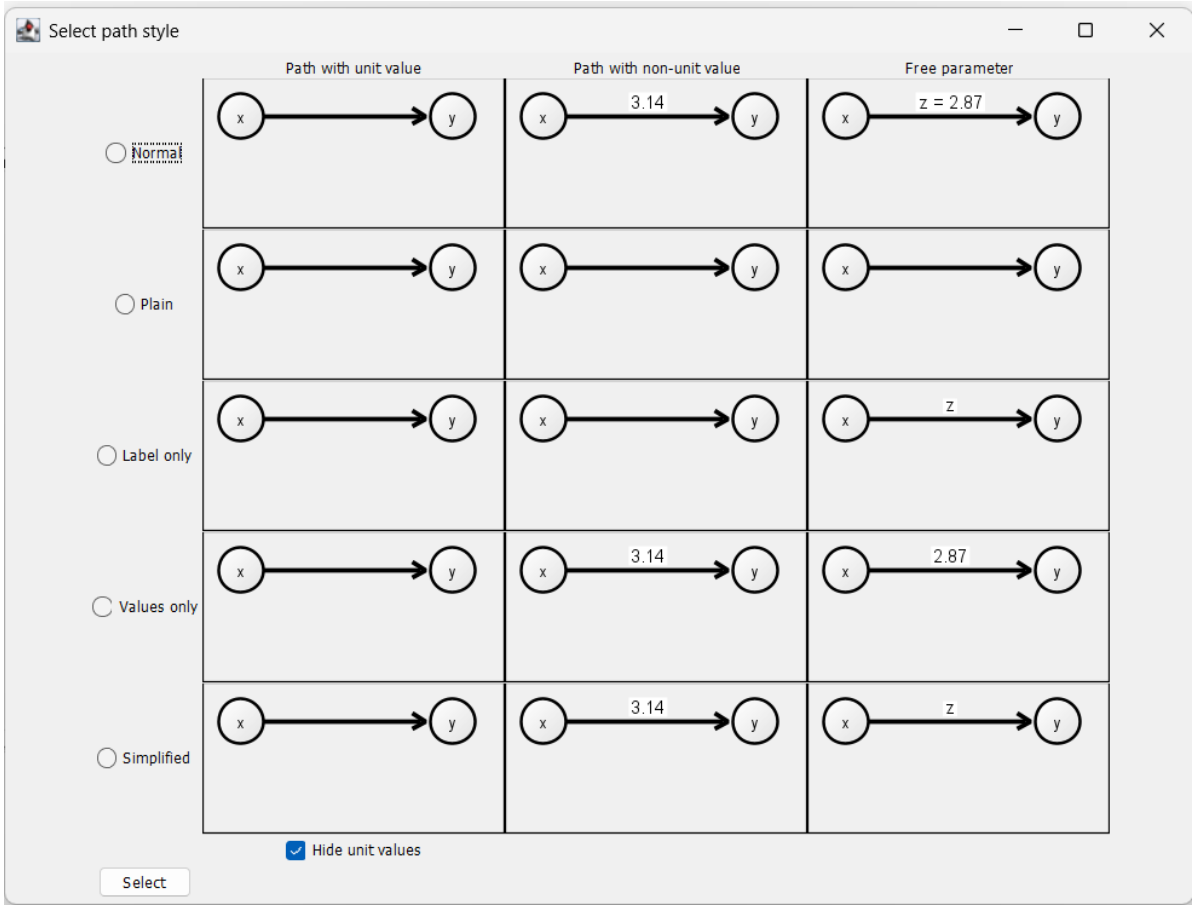


Here are three examples of styles:



2.4 Path Styles

By default, Onyx shows path with a value fixed to one without any label. Paths that have a value fixed but different than one are shown with the value as label. Freely estimated parameters are shown with both their label and their value (e.g., "z = 2.87"). You can also choose different path styles by right-clicking on a model, then selecting "Customize Model -> Change Path Style". This opens the following window in which you can select among several alternative options. In particular, the option "Simplified" may be useful to create figures for manuscripts.



2.5 Exercises

- Create a factor model with one latent factors, four observed variables, residual variances for each factor, a latent factor variance, and factor loadings; fix the first factor loading at "1" and let the others be freely estimated; also freely estimate all residual variances; rename the observed variables to "item1", "item2", ...; rename the factor to "factor", rename the factor loadings to lambda1, lambda2, ... but use greek letters.
- Apply diagram styles and make your own modifications
- Export the diagram as image file

3 Goodness of Fit

3.1 Model Fit

In SEM, the fit function to obtain maximum likelihood estimates under multivariate normality of the observations and assuming no missing data is:

$$F_{\mathrm{ML}} = \log|\Sigma(\theta)| + \mathrm{tr}\left(\mathbf{S}\,\Sigma(\theta)^{-1}\right) - \log|\mathbf{S}| - p$$

Assuming H_0 that there is no misfit is true, the following test statistic has an asymptotic χ^2 distribution

$$\chi^2 = (N-1)\,F_{\mathrm{ML}}$$

3.2 Model Fit

- χ^2 is the fundamental model fit measure as discrepancy between model-implied (co)variances and means and observed (co)variances and means
- Significance test of close fit is usually not recommended as it has very high power; instead we usually favor effect-size-type metrics
- CFI: discrepancy between an independence model and the hypothesized model; *CFI asks: How much better does my model fit compared to a model that assumes no relationships at all?*
- RMSEA: estimates how much misfit per degree of freedom your model has in the population. *RMSEA asks: "Is the model achieving good fit without overusing parameters?*

3.3 Residuals

Residuals represent the difference between what the model predicts and what is actually observed in the data. More specifically, SEM works by reproducing the observed covariance (or correlation) matrix using a model-implied covariance matrix. Residuals are simply the element-wise differences between these two matrices. If the model fits well, these residuals should be small—ideally close to zero—because the model is successfully capturing the relationships among variables.

Residuals are useful because they provide localized information about model misfit. Global fit indices like chi-square, CFI, or RMSEA tell you whether the model fits overall, but they don't tell you where it fails. Residuals, in contrast, point to specific pairs of variables where the model does a poor job reproducing the observed relationship. For example, if two indicators have a large positive residual, it means their observed correlation is higher than what the model predicts, suggesting that something is missing, such as a residual correlation or an additional factor.

3.4 Raw and Standarized Residuals

There are different types of residuals in SEM. Raw residuals are simple differences between observed and implied covariances, while standardized residuals scale these differences by their expected variability, making them easier to interpret across different variable pairs. Standardized residuals are often preferred because they help identify which discrepancies are unusually large relative to sampling variability.

Importantly, residuals are not just diagnostic tools – they also guide model refinement. Large residuals can suggest theoretically meaningful modifications, such as allowing two indicators to correlate due to shared wording or content (e.g., shared method variance). However, they must be interpreted carefully: adding parameters purely to eliminate residuals can lead to overfitting and models that do not generalize well.

3.5 Model-implied and observed covariances

A summary measure of the misfit in the residuals is the standardized root mean square residuals, which is:

$$\text{SRMR} = \sqrt{\frac{2}{p(p+1)} \sum_{i \leq j} \left(r_{ij} - \hat{r}_{ij}\right)^2}$$

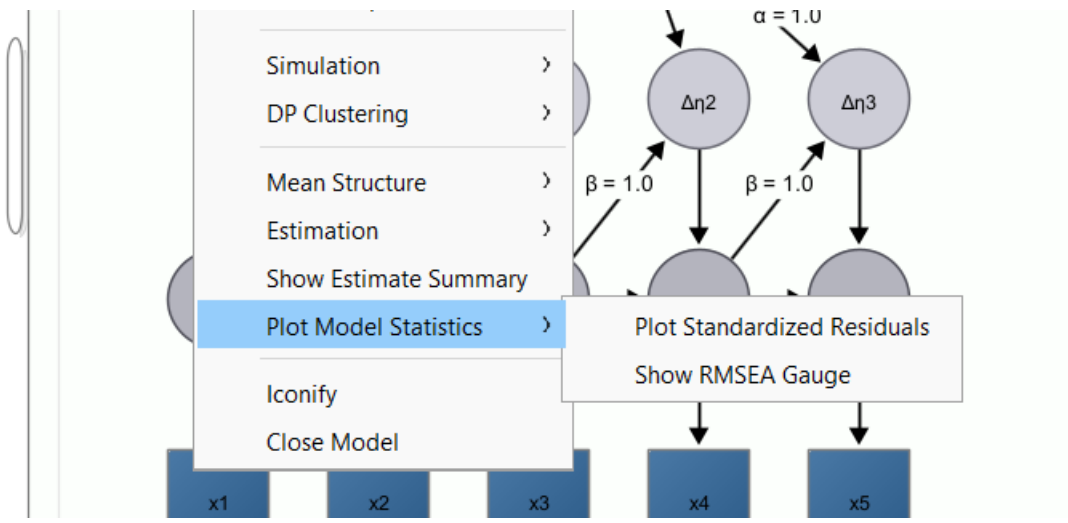
where

r_{ij} are the observed correlations $\hat{r_{ij}}$ are the model-implied correlations

The exact computation differs across programs. Some programs standardize such that the means are included and some don't. Some programs standardize such that the diagonal elements are included and some don't. Note that Onyx standardizes the observed covariance matrix and then the model-implied covariance matrix each with their own variances. Thus, Onyx SRMR does not include misfit in the means or the residual variances – just the covariances.

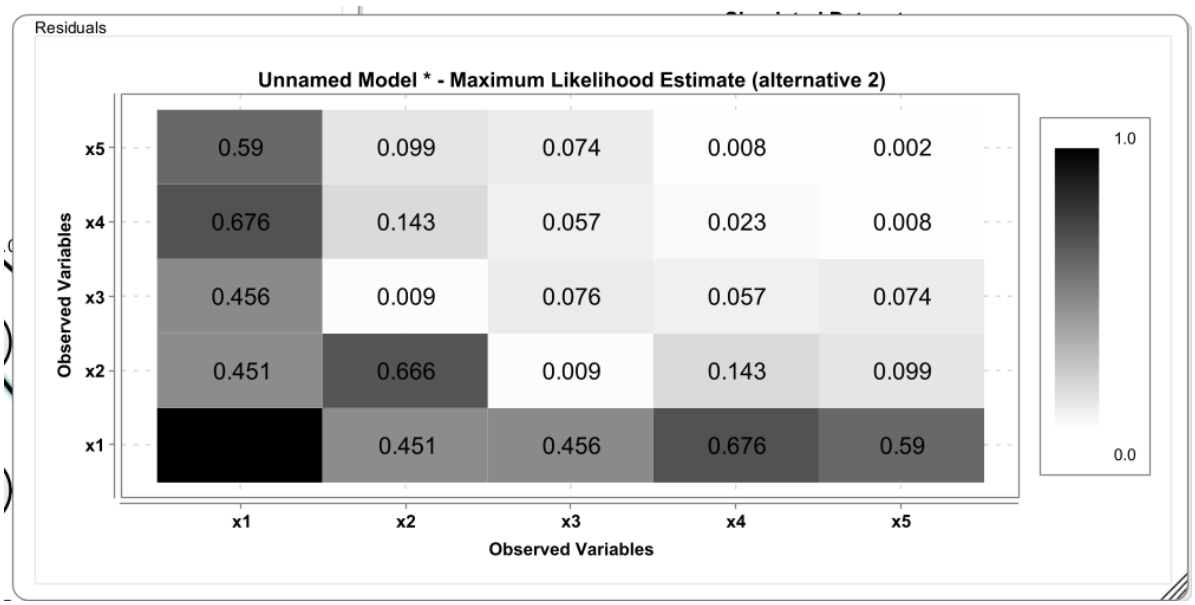
3.6 Model Fit Plots

There are two new plotting features to obtain visual information about model fit. Right-click on a model and select "Plot model statistics". Then choose "Plot Standardized Residuals"



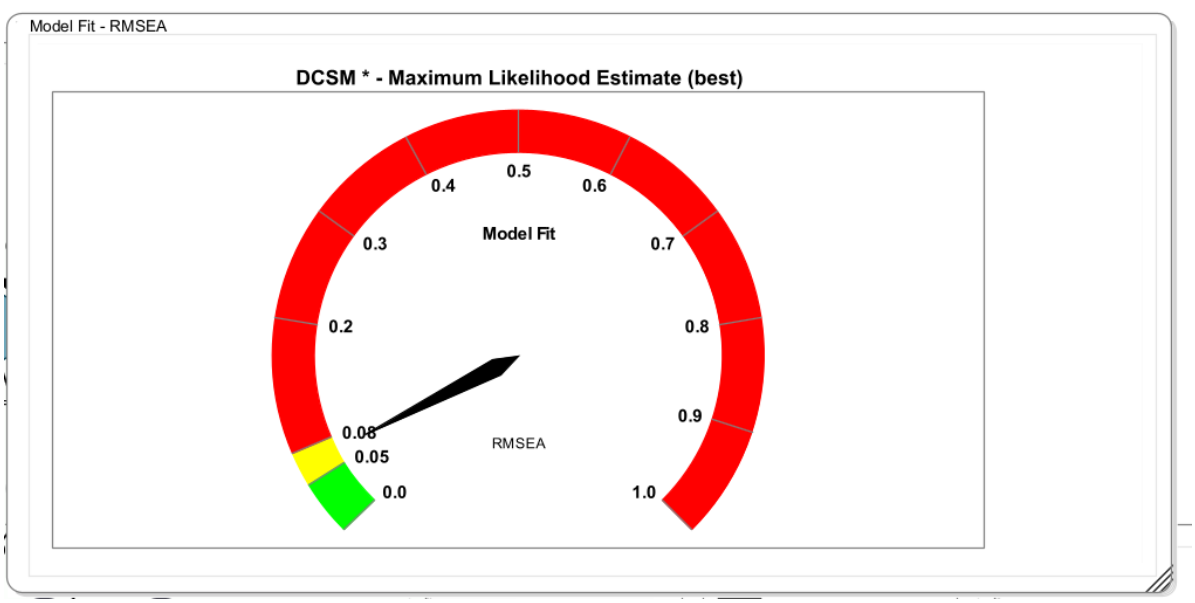
3.7 Residual Plots

This opens the standardized residuals plot. This plot show standardized residuals that are slightly different than what is used to compute the SRMR. These standardized residuals are computed such that they also represent misfit in the diagonal of the matrix, that is, the residual variances. This is achieved by standardizing both model-implied covariances and observed covariances with the variances of the observed covariance matrix.



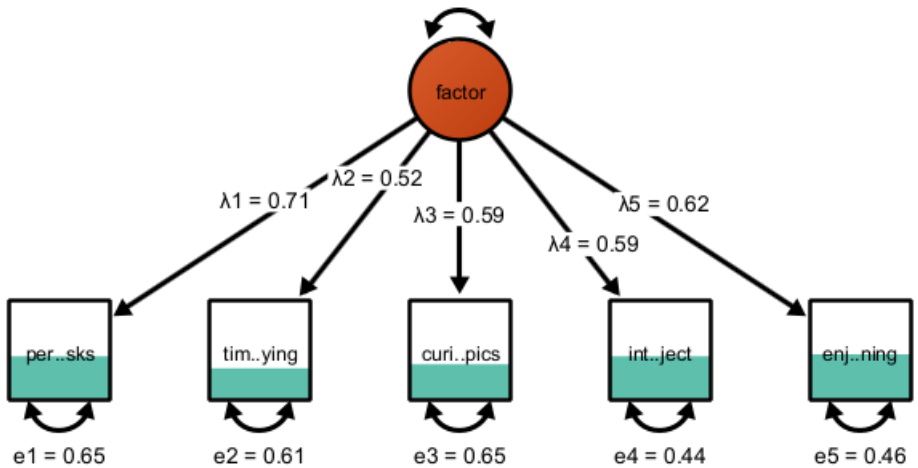
3.8 Model Fit Gauge

A further visualization of model misfit is the RMSEA gauge. This is a gauge that shows the RMSEA and shades some region of the RMSEA according to whether those ranges were proposed as ranges of acceptable or good fit.



3.9 Visualizing Explained Variance

For observed variables, you can select a fill-style that reflects the percentage of explained variance in a variable. Right-click on a variable or a selection of variables, click "Customize Variable", then "Select Fill Style", then "Bokers R2". This applies a fill-style that fills the selected variable only up to the level that corresponds to the variance explained by other variables in the model. The unexplained part is not filled. This gives a good quick overview, for example in factor models, how well an item works. Here is an example in a factor model with five indicators:

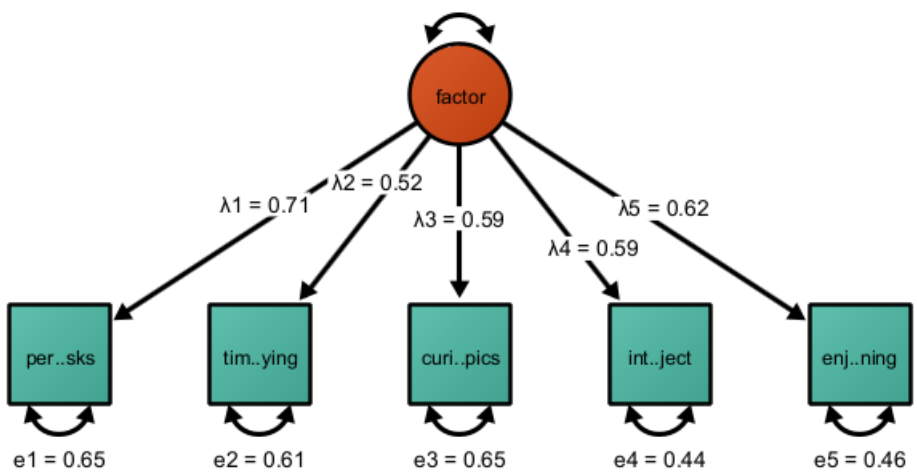


3.10 Exercise

- Open the dataset "academicmotivation.csv" in Onyx
- This is a simulated dataset of a common factor of academic motivation that is measured with five indicators: persistence_on_tasks, time_spent_studying, curiosity_about_topics, interest_in_subject, and enjoyment_of_learning
- Fit a common factor model (just covariances; no means); apply Boker's R2 style to the observed variables
- Inspect model fit, model-implied covariances, and residuals (using the plot)
- Given these statistics and the information that interest_in_subject and and enjoyment_of_learningtion tap more into intrinsic_motivation than the others, how could we modify the model?

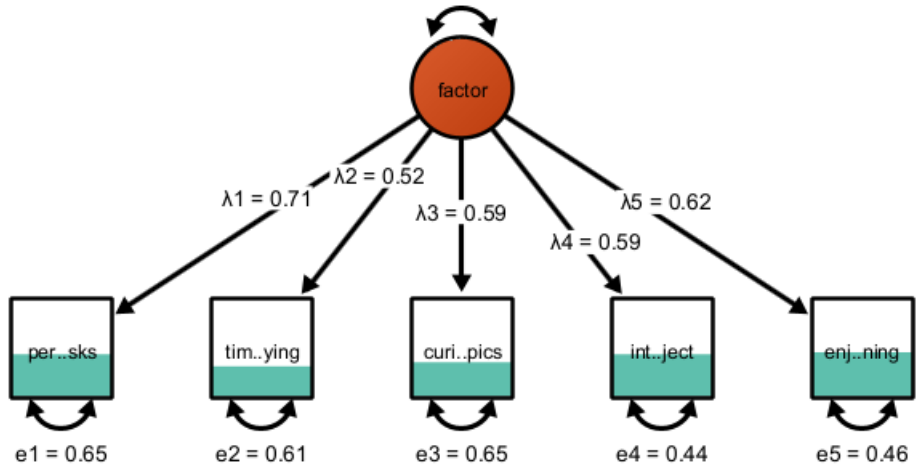
4 Solution

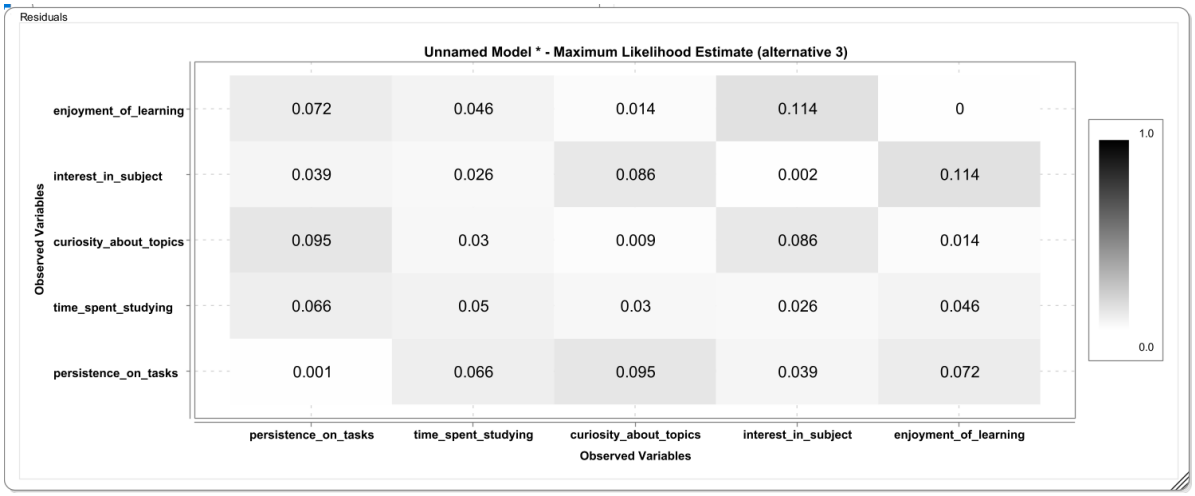
4.1 Factor Model



Model fit is $\chi^2 = 21.033$, RMSEA=0.106, CFI=0.106, SRMR=0.055. RMSEA and CFI are outside regions that are usually deemed acceptable.

Applying Boker's R2 style, this yiels:





The large discrepancy in the correlation of enjoyment_of_learning and interest_in_subject may be an indication that these items also capture something different than the other items. In light of a theory (intrinsic vs academic motivation), this can be used to modify the factor model to include a residual correlation (which is akin to a second common source of variance) to the model. By adding another free parameter, model misfit in the present sample is reduced. Note that such model modifications after having seen the data should always be reported as exploratory.

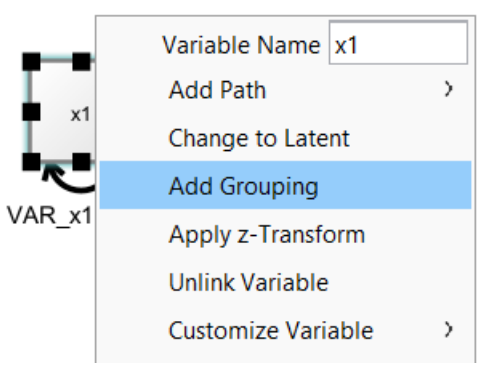
5 Multi-Group Models

Multi-group models in Structural Equation Modeling (SEM) are an extension of standard SEM techniques that allow researchers to test whether relationships between variables are consistent or different across two or more groups. In the context of Psychology, these groups might represent categories such as gender, age cohorts, cultural backgrounds, or clinical vs. non-clinical populations. By estimating the same model simultaneously in multiple groups, researchers can formally assess measurement invariance (whether constructs are interpreted similarly across groups) and structural invariance (whether the relationships between constructs differ). This is especially valuable because it ensures that observed group differences reflect true psychological distinctions rather than artifacts of measurement bias. Overall, multi-group SEM provides a rigorous framework for comparing theoretical models across populations, strengthening the validity and generalizability of psychological findings.

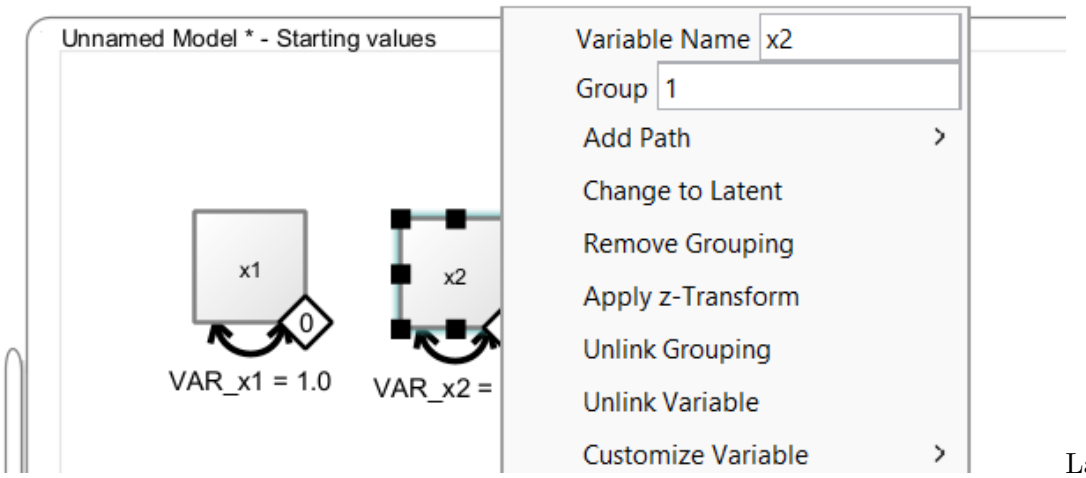
5.1 Multi-group models in Onyx

Onyx support multi-group models, in which we use a grouping variable that encodes group membership between individuals. Onyx has its own approach to multigroup modeling which works with filters on observed variables that filter out subsets of observations that should be modeled by a given observed variable.

By right-clicking on an observed variable, you can select "Add Grouping" to add a filter to one or more selected observed variables.



Then, a little diamond grouping indicator appears on that variable. The value inside the diamond reflects the group membership. To change group membership, right-click the diamong and change the "Group" value:



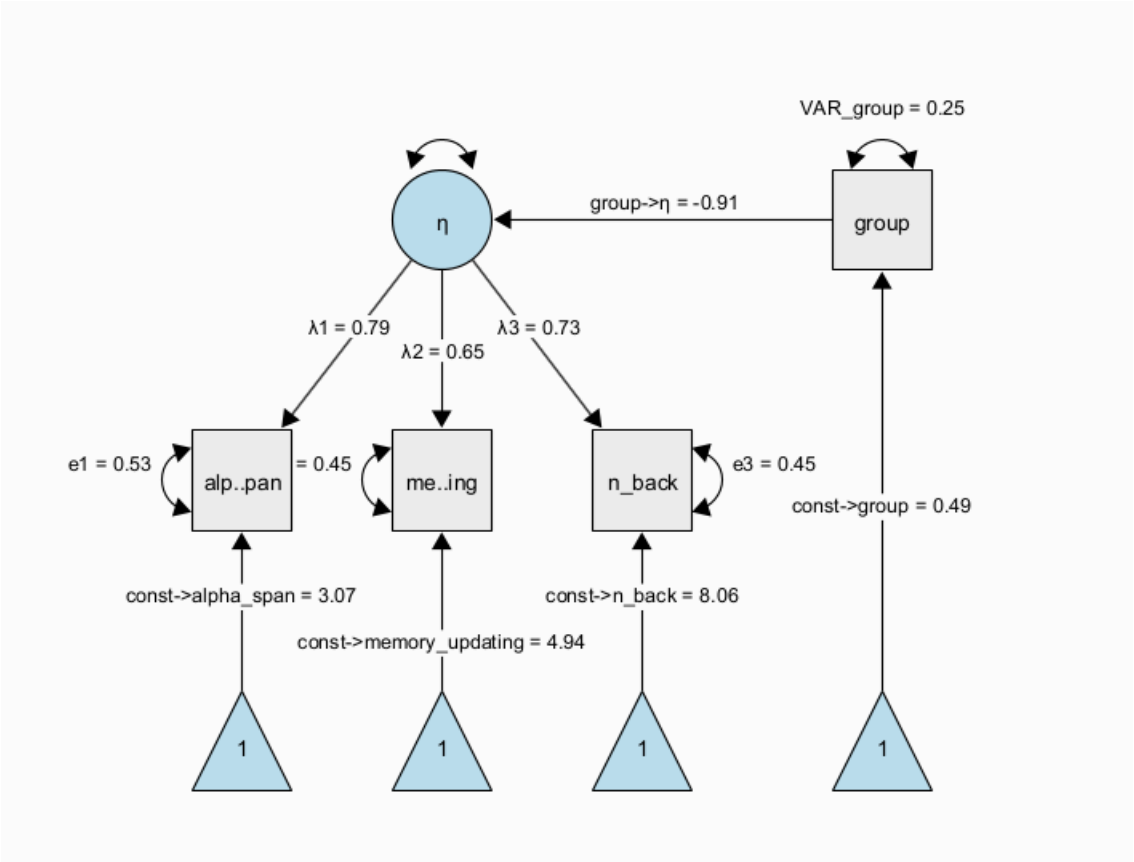
Latent Mean Differences

The diamonds are variable containers, that is, you can drag a discrete-valued grouping variable from a dataset on to the diamond. As long as the diamond is not filled (white fill color), it is not connected. Once it is connected, it becomes filled (black color), that is, it is now associated with a grouping variable in the dataset.

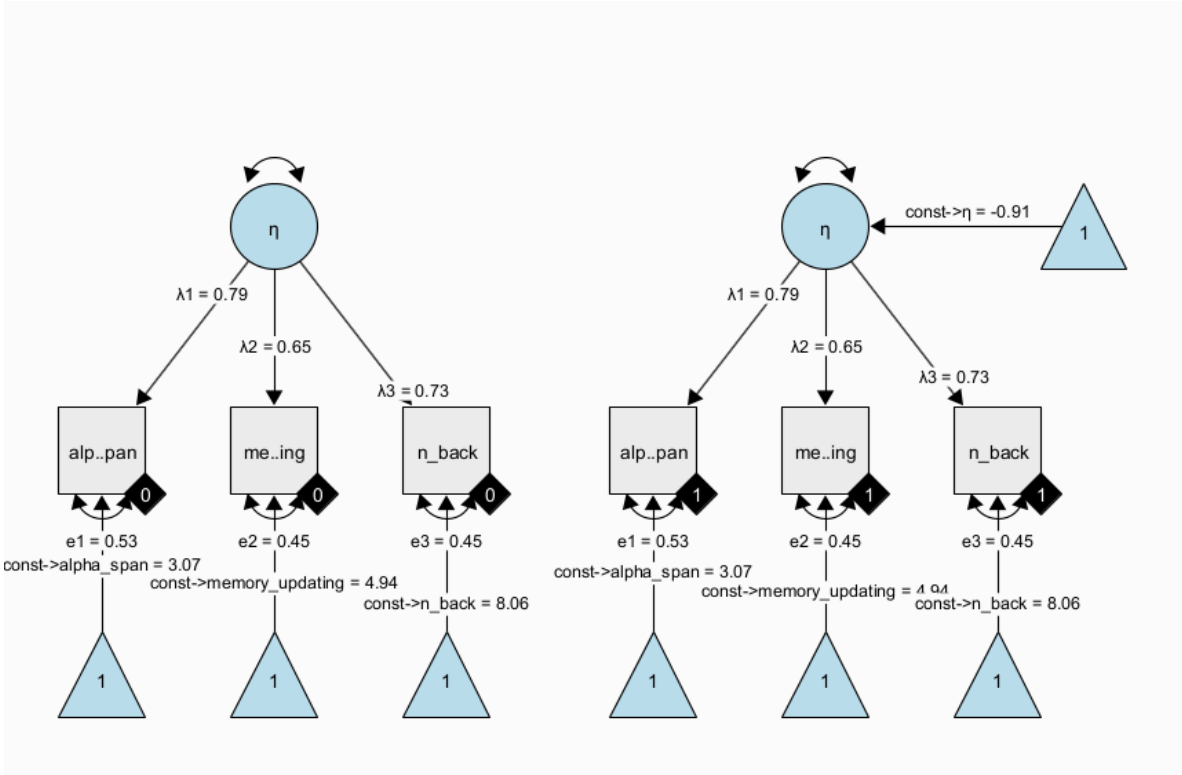
6 Applied Example: Latent Means

Models of latent mean differences are like the classical t-test or ANOVA but estimated on latent constructs.

Figure 1 shows a path diagram representing one possibility to set up a latent mean difference model when there are multiple, noisy measurements of the same underlying latent construct. This model is a MIMIC (multiple-indicators multiple-causes, **hancock2001effect?**) model, in which we estimate a measurement model of a latent trait η across two groups that are encoded via a dummy group indicator. The latent trait is measured through multiple observed indicators (here, $x1$ to $x3$) with indicator-specific residual variance terms, which allows to separate error variance from true-score variance. If correctly specified, this approach allows us to obtain more precise estimates of the latent trait purged of measurement error and unreliability (DeShon 1998). In this specific instance, the regression path from the dummy-coded group indicator to the latent variable directly represents a standardized mean difference because the scale of the latent variable was fixed to have a variance of one. Note that the MIMIC model assumes that measurement invariance holds, that is, the factor loadings, item intercepts, and item residual variances are assumed to be the same across groups.



A more flexible model specification is also possible; using a multiple-group factor model, we can estimate the same factor model in each group separately and empirically test whether measurement invariance holds; then, we can continue to test whether there is a difference in latent factor means across groups. This is also known as the SMM [structured means modeling; (**hancock2001effect?**)] approach:



Note that all parameters with the same name have identity constraints; in particular, this means that parameters with the same name are identical across groups. In the path diagram shown above, all parameters are identical across groups. This implements a very strong assumption of measurement invariance across groups and identity of magnitude of variance (= individual differences at the latent level) across groups. Again, this restricted version of the SMM model is identical to the above dummy-coded MIMIC approach. However, the SMM allows to also freely estimate parameters between groups and perform tests of whether some of the restrictions are supported by the data or not.

6.1 Exercises

- Load the dataset "factor_model_memory.csv"
- Use the "Latent Means Model" wizard to create both the SMM and the MIMIC approach model to estimate models of latent mean differences
- Freely estimate the latent variance in the second group of the SMM approach
- Copy the SMM model and freely estimate the factor loadings across groups; run a likelihood ratio test between the models as a test of measurement invariance

7 Model Comparison

7.1 Model Comparison

Model comparison in Structural Equation Modeling is a central technique for evaluating how well different theoretical specifications fit the observed data. In Psychology, researchers often compare **nested models**, where one model is a constrained version of another (e.g., fixing certain parameters to zero or imposing equality constraints). This nesting allows for the use of **likelihood ratio tests (LRTs)**, which formally assess whether the more complex model provides a significantly better fit than the simpler one. By examining changes in model fit (typically via differences in chi-square statistics for nested models) researchers can determine whether added parameters meaningfully improve explanatory power or whether a more parsimonious model is sufficient. This approach is particularly useful for testing theoretical assumptions, such as whether specific pathways are necessary or whether constraints (e.g., equal effects across groups or time points, measurement invariance, additional predictors) are justified, thereby supporting more precise and theory-driven conclusions.

7.2 Model Comparison using Chi-Square-Tests

Onyx supports model comparison for nested models based on the chi-square test. To compare two models, you can add a model comparison of two models that is represented as an edge between two models (similar to a path between two variables). The model comparison is created analogously to creating a path between two variables. Press the right mouse button on an empty space of one model, the mouse pointer onto the second model, and release the mouse button. A model comparison edge will appear between the two models:

7.3 Holzinger Swineford

From the description provided in the lavaan package:

> The classic Holzinger and Swineford (1939) dataset consists of mental ability test scores of seventh- and eighth-grade children from two different schools (Pasteur and Grant-White). In the original dataset (available in the `MBESS` package), there are scores for 26 tests. However, a smaller subset with 9 variables is more widely

used in the literature (for example in Joreskog's 1969 paper, which also uses the 145 subjects from the Grant-White school only).

The commonly used factor model for these 9 variables consists of three latent variables each measured by three indicators:

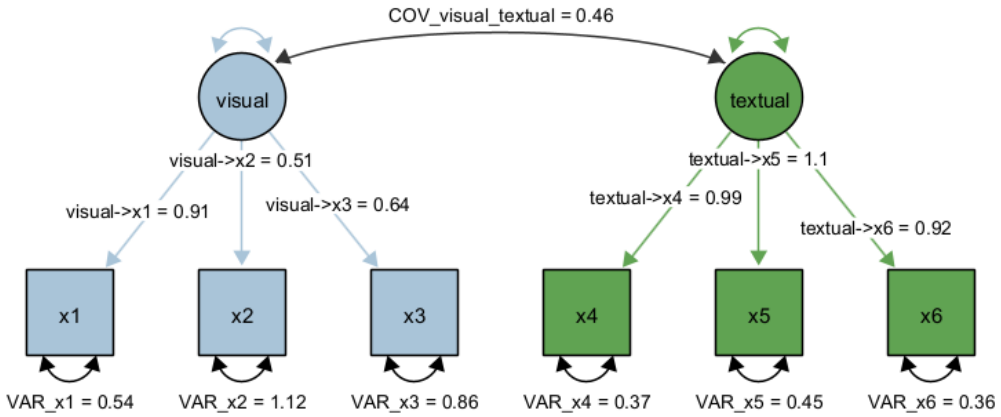
- a *visual* factor measured by three variables: `x1`, `x2` and `x3`
- a *textual* factor measured by three variables: `x4`, `x5` and `x6`
- a *speed* factor measured by three variables: `x7`, `x8` and `x9`

7.4 Exercise

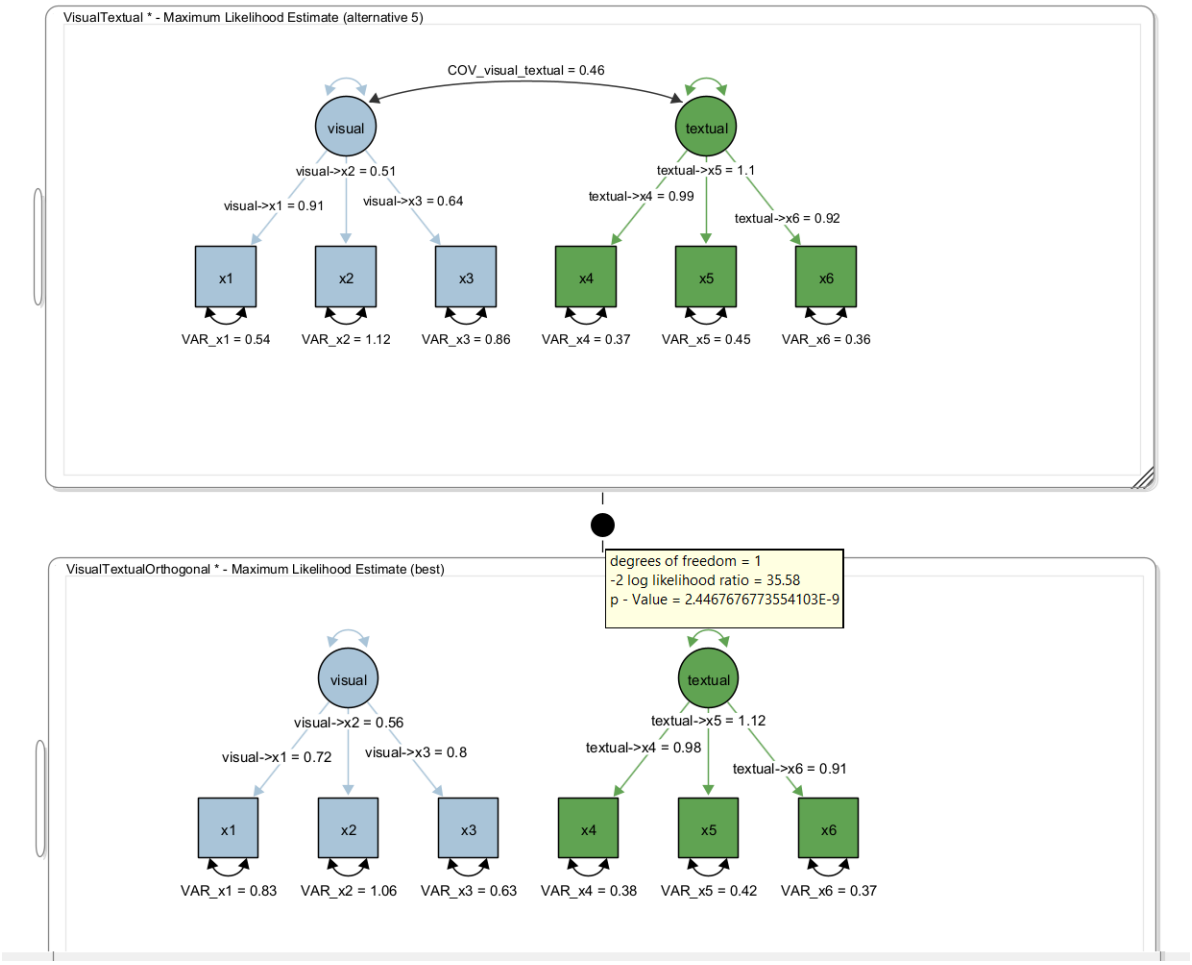
1. Load the dataset `HolzingerSwineford1939.csv`
2. Create a latent factor model with the first six observed variables (x1 to x6); allow a correlation of the two latent factors; name the model "VisualTextual"
3. Make a copy of the model ("edit -> clone model") and rename it to "VisualTextualIndependent"
4. Modify the model "VisualTextualIndependent" by removing the correlation between the two latent factors
5. Use a likelihood ratio test to compare the two models. Which model is favored? Is there evidence for a correlation of the textual and visual factor?
6. Obtain latent variable factor scores and plot them in a scatterplot

7.5 Solution

First, we load the dataset, we select variables x1 to x6 (e.g., by selecting x1, holding down the SHIFT key and then selecting x6 in the variable list of the dataset view). Next, we create two latent factors and draw the factor loadings. Last, we fix the latent variances to one and estimate all factor loadings freely; alternatively, we could fix one factor loading to one and estimate the latent variances freely. However, by fixing the latent variances to one, we can directly interpret the latent covariance as correlation. The path diagram should look like this:



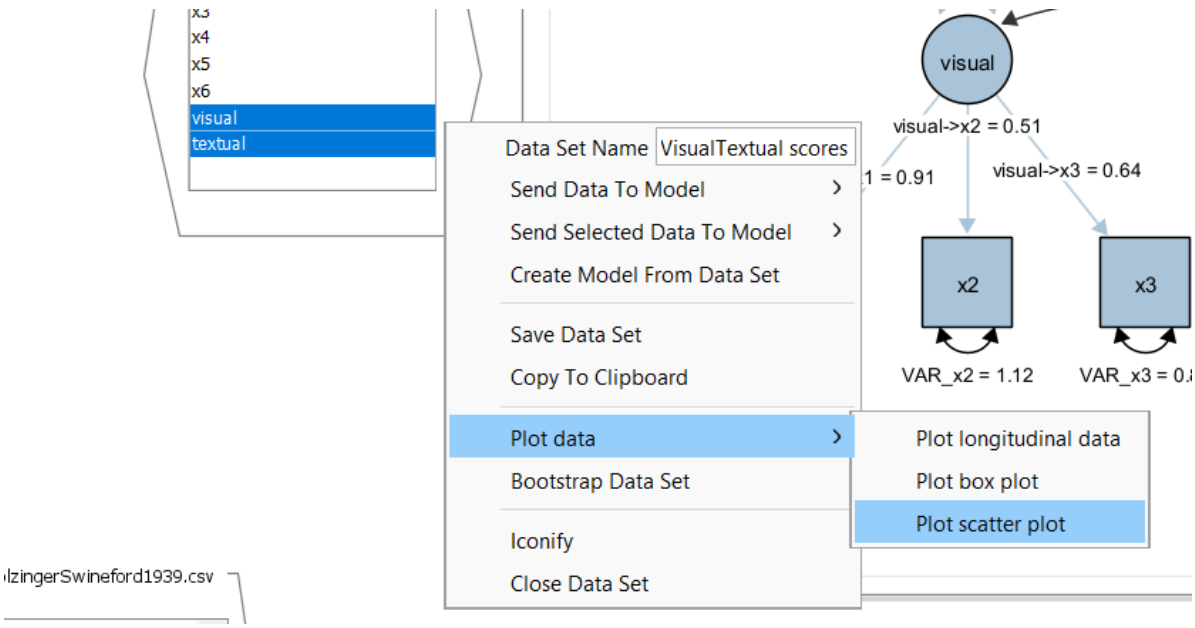
Next, we clone the model (using Edit -> Clone Model) and rename the cloned model. Then we right-click the covariance in the cloned model and select "Delete Path". Then, create a model comparisong edge between the two models. Hover the mouse over the filled circle of the model comparison edge and the result of the model comparison appears:



Here, we obtained a test statistic of 35.58 for the likelihood ratio test with one degree-of-freedom (because a single parameter was removed, ie., set to zero). The p value is <0.001 and thus, the model with the extra parameter (ie., the covariance) significantly fits better than the model without the covariance.

8 Factor Scores

Factor score estimation is a post-estimation step in SEM that involves deriving individual-level scores on latent variables from the fitted model. Ideally, any kind of inference should be made using the SEM instead on estimated individual-level scores because the estimation introduces estimation error. For example, if one is interested in the correlation of two latent factors, this correlation should be modeled and model-based estimates should be used for the inference. Common approaches include regression (or "Thurstone") scoring, Bartlett scoring, and empirical Bayes methods, each trading-off bias and efficiency. Importantly, factor scores are not uniquely determined—they depend on the chosen estimation method and model specification—so they should be interpreted as approximations rather than exact measurements of the latent constructs. In applied work, factor scores are often used for secondary analyses, such as predicting outcomes or classifying individuals, but their uncertainty and potential indeterminacy should always be taken into account.



8.1 Factor-score estimation in Onyx

8.2 Holzinger Swineford

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The commonly used factor model for these 9 variables consists of three latent variables each measured by three indicators:

- a visual factor measured by three variables: x1, x2 and x3
- a textual factor measured by three variables: x4, x5 and x6
- a speed factor measured by three variables: x7, x8 and x9

8.3 Exercise

1. Load the dataset `HolzingerSwineford1939.csv`
2. Create a latent factor model with the first six observed variables (x1 to x6); allow a correlation of the two latent factors; name the model "VisualTextual"
3. Obtain factor scores of the two latent variables
4. Plot a scatter plot of the estimated factor scores from the visual and textual factor. What do you see? How does this relate to the latent correlation between the two factors?